

CITYTWIN · DESIGN DECISIONS REGISTER

CityTwin design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CTW-DEC-001	Version	0.4
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/citytwin		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Design for construction (CTW-DDR-003, P1 to P14) accepted by Amish on 2026-10-02; moved from open for review to decisions made
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 7 on 2026-10-02 (shield holes only where fitted, sheltered pilot near Irving, antenna walk test, languages, R20 kept as post-pilot target, pull format, CrossSafe uplink request); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

CityTwin design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`, CTW-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

Table 2. Items to confirm against the chosen part.

#	What to confirm	Why it matters	Source
1	The cabinet comes with a removable mounting plate of about 340 x 390 mm on studs that stand it at least 10 mm off the back wall, and a lockable door on the 360 x 460 mm face	The rails, the cabinet screws behind the plate and the 25 mm cable hole above the plate are placed for this	CTW-DDR-003, P9
2	The e-paper panel's outline (210 x 290 x 7 mm assumed) and where its flat cable leaves the panel	Sets the foam gasket, the carrier's ribbon slot and the driver board position	CTW-DDR-003, P4
3	The M8 steel flush rivet nuts' grip range covers the 4 mm post wall with its coating, and their countersink size	The head and cabinet screws depend on them	CTW-DDR-003, P1
4	The buttons' LED rings run from 5 V or 12 V, and their nut size fits beside the notice plate	Sets the LED driver wiring and the clearance below the notice plate	CTW-DDR-003, P5, P6
5	TwinKit fits a backup pack of at least 1.84 Ah for cabinet installs (now 1.5 Ah); until then R18 is not met in the worst case	The outage ride-through of 2 h	CTW-DDR-002, N3
6	The hood light's LED strip and dusk sensor work from the 12 V supply through the LED driver, dimmed to about 0.3 W	Night lighting (R9) and running power (R15)	CTW-DDR-002, N6
7	The anchor design resistance (10 kN per M16 anchor assumed) and the footing, from a structural design to the local code	Wind (R13) and the safety of a kiosk of about 64 kg beside a walkway	CTW-CAL-001, section H

■ Value engineering

Value-engineering target: USD 750 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 778 for the pilot kiosk (USD 28 over the target; the pilot site is sheltered, so no sun shield or frost heater, decided 2026-10-02); USD 808 with the outdoor sun shield, and USD 1,068 with the TwinKit gateway, which is costed in TwinKit.

- **Main cost drivers:** the parts added to make the kiosk buildable (display carrier USD 8, fixings USD 22, and USD 13 more for the head's converter, LED driver and terminal block, less USD 4 moved out of another item); the outdoor sun shield (USD 30), reported separately.

- **Savings worth trying:** a lower-cost cabinet; dropping the real-time clock; checking prices at purchase, since the figures are indicative.

■ Decisions made

Table 3. Decisions made, with the record that argues each one.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D11: budget covers the kiosk only; color e-paper with an indoor LCD variant; sheltered pilot; gateway in the kiosk cabinet; mains power; 5 per hour small-count threshold and two years of raw data; CC BY 4.0; buttons; 46-node reference neighborhood; one-way publishing; outbound pulls for PotholeLog and DockHub	Amish: "i accept all your recommendations, go with them across all repos."	CTW- DDR-001, CTW- DDR-002
2026-09-25	N1 to N6: budget_usd \$750; R10 restated (LED ring within 0.5 s, layer within 25 s); larger TwinKit pack requested; cabinet sun shield and shaded screen as outdoor site rules, heater only for a frost city; pilot range 0 to 35 °C air; hood light dimmed to about 0.3 W	Amish, same instruction	CTW- DDR-002
2026-10-01	Design for construction, P1 to P14: rivet nuts in the post for the head and cabinet; folded head tray with a removable front panel; window bonded with glazing tape; display carrier on studs; 5 V converter, LED driver and terminal block in the head; button, notice plate and hood fixings; mounting plate in the cabinet; grommets cable holes; conduit hole and gland; vent hole for galvanizing; antenna bracket plate; shield on standoffs with a thumb-screw back panel	Made under Amish's 2026-09-30 instruction to make the design physically buildable ("If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."); the changes themselves were accepted on 2026-10-02 (below)	CTW- DDR-003
2026-10-02	Sun shield standoff holes: option (a), drilled only in cabinets that get the shield, so the sheltered pilot cabinet keeps its unbroken IP55 sides	Amish: "i approve your recommendations for all 555 open decisions."	CTW- DDR-003, A2
2026-10-02	First partner and pilot site: a city within easy reach of Irving, Texas, with an active open-data program, the kiosk in a sheltered public lobby such as a library or transit center; with a sheltered site the frost heater is not fitted. The city is chosen as the first candidate to approach, not yet agreed	Amish: "i approve your recommendations for all 555 open decisions."	CTW- DDR-001, O1; CTW- DDR-002
2026-10-02	Antenna: keep the post-top antenna at 2.38 m for the pilot and run a coverage walk test from the site at TRL 4; if nodes in the reference neighborhood fall short, move to a rooftop gateway, not a taller mast	Amish: "i approve your recommendations for all 555 open decisions."	CTW- DDR-002, Q1
2026-10-02	Languages: English plus the partner city's most widely spoken other language in its own script, with a QR code and a phone or text number on the notice plate for questions and fault reports; refined with the partner in co-design	Amish: "i approve your recommendations for all 555 open decisions."	CTW- DDR-002, Q2
2026-10-02	Street climate (R20): kept as a post-pilot target; outdoor sites only in permanent shade with the sun	Amish: "i approve your recommendations for all 555 open decisions."	CTW- PRC-001, open

Date	Decision	Decided by	Record
	shield; a panel rated above 40 °C is sought at TRL 4, and the heated bay is fitted only for a frost city		questions; CTW-REQ-001, R20
2026-10-02	Pull paths: the outbound pull already decided (D11), with daily CSV files on a fixed HTTPS address at each operator, a published column list and a version field; the same format to be agreed in PotholeLog (its open item 6)	Amish: "i approve your recommendations for all 555 open decisions."	CTW-DDR-001, D11
2026-10-02	CrossSafe uplink: ask CrossSafe to add an hourly LoRaWAN summary uplink from one side of the crossing; CityTwin shows no CrossSafe layer until it exists	Amish: "i approve your recommendations for all 555 open decisions."	CTW-REQ-001, R1
2026-10-02	Design for construction accepted: the changes P1 to P14, as made	Amish: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)"	CTW-DDR-003, Table 1